Parameter change files

Purpose

`cal_parms.cal` is the dictionary of every parameter that calibration can touch: its name, the object type it lives on, and the absolute range it may be moved through. `calibration.cal` is the change list applied during a hard-calibration run: each row picks a parameter from the dictionary, sets a change type and magnitude, optionally restricts the change to a list of objects, and optionally attaches conditions. Both files feed the same parameter selector (`cal_parm_select`) used by hard calibration and by parameter changes inside soft calibration.

Source

- `cal_parms.cal` reader: src/cal_parm_read.f90
- `calibration.cal` reader: src/cal_parmchg_read.f90
- Type definitions: src/calibration_data_module.f90, `type calibration_parameters` (rows of `cal_parms.cal`) and `type update_parameters` (rows of `calibration.cal`).
- File slot names: src/input_file_module.f90, `type input_chg`. Defaults: `cal_parms = "cal_parms.cal"`, `cal_upd = "calibration.cal"`.

cal_parms.cal

`cal_parm_read.f90` reads:

- Line 1: title (skipped).
- Line 2: integer `mchg_par`, the number of parameter records that follow.
- Line 3: header (skipped).
- Lines 4 .. 3 + `mchg_par`: one record per parameter, read list-directed into `cal_parms(i)`.

Record layout (`type calibration_parameters`):

#	Field	Type	Description
1	`name`	char(25)	Parameter name. Must match a case in `cal_parm_select.f90`. Examples: `cn2`, `awc`, `alpha`, `epco`, `lat_ttime`.
2	`ob_typ`	char(25)	Object type the parameter is attached to: `hru`, `hlt`, `sol`, `lyr`, `plt`, `aqu`, `cha`, `sdc`, `rte`, `swq`, `res`, `rdt`, `ru`, `bsn`, `pcp`, `tmp`, `gwf`. Determines which collection is iterated.
3	`absmin`	real	Lower bound. Any change is clamped to this value.
4	`absmax`	real	Upper bound. Any change is clamped to this value.
5	`units`	char(25)	Units string for the parameter (`mm`, `m`, `days`, `null`, …). Documentation only; the reader does not interpret it.

Example

`refdata/Osu_1hru/cal_parms.cal`:

```
cal_parms.cal: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
186
name                        obj_typ       abs_min       abs_max             units
cn2                             hru      35.00000      95.00000              null
cn3_swf                         hru       0.00000       1.00000              null
```

```
usle_p                          hru         0.00000          1.00000              null
ovn                             hru         0.01000         30.00000              null
elev                            hru         0.00000       5000.00000                 m
slope                           hru         0.00010          0.90000               m/m
slope_len                       hru        10.00000        150.00000                 m
lat_ttime                       hru         0.50000        180.00000              days
```

The count line on line 2 (`186`) drives the allocation of `cal_parms(0:mchg_par)`. The header on line 3 (`name`, `obj_typ`, `abs_min`, `abs_max`, `units`) is skipped; values are read by position.

calibration.cal

`cal_parmchg_read.f90` reads:

- Line 1: title (skipped).
- Line 2: integer `mcal`, the number of change records.
- Line 3: header (skipped).
- Then `mcal` change records, each followed by `conds` condition lines (often zero).

Change record

Each change record is one logical line, read list-directed in two passes. The first pass reads the fixed prefix; if the prefix indicates an object list is present, the line is backspaced and re-read to pick up the integers.

Fixed prefix:

#	Field	Type	Description
1	`name`	char(25)	Parameter name. Must match a `name` in `cal_parms.cal`. The reader crosswalks this to `num_db`.
2	`chg_typ`	char(16)	Change operator. One of `absval` (replace), `abschg` (add), `pctchg` (percent change).
3	`val`	real	Magnitude of the change, interpreted per `chg_typ`.
4	`conds`	int	Number of condition lines that follow this record. May be zero.
5	`lyr1`	int	First soil layer in range (0 means all layers). For reservoir decision tables, reused as a storage-zone flag.
6	`lyr2`	int	Last soil layer in range (0 means through last layer).
7	`year1`	int	First year (for climate parameter changes).
8	`year2`	int	Last year (for climate parameter changes).
9	`day1`	int	First day of year (for climate parameter changes).
10	`day2`	int	Last day of year (for climate parameter changes).
11	`nspu`	int	Number of object indices listed after this prefix. `0` means apply to every object of this parameter's `ob_typ`.

If `nspu > 0`, the same line continues with `nspu` integers giving the object indices to update. When `nspu == 0` the reader allocates a `num` array that lists every object of the matching type (every HRU, every channel, every plant, and so on) so the change applies globally.

Condition lines

A condition line takes one of two forms, distinguished by the literal token in column 1:

- `range <var> <val1> <val2>`. Restricts the change to objects whose `<var>` falls between `val1` and `val2`. Used in `cal_conditions.f90` for `res_pvol`.
- `<var> <alt> <targ> <targc>`. Read as a `calibration_conditions` record. Used for categorical filters: `hsg`, `texture`, `plant`, `pl_class`, `landuse`, `cal_group`, `res_typ`.

See Conditions and regionalization for the supported variable list.

Example

`refdata/Osu_1hru/calibration.cal`:

```
  hru-new.cal developed from soft data calibration
    13
 NAME           CHG_TYP                  VAL    CONDS  LYR1  LYR2   YEAR1  YEAR2   DAY1   DAY2  OBJ_TOT
cn2         pctchg                 7.714000E+00  0          0          0          0          0
petco       pctchg                -5.273950E+01  0          0          0          0          0
latq_co     pctchg                -3.732030E+01  0          0          0          0          0
alpha       pctchg                 6.859000E+00  0          0          0          0          0
epco        pctchg                 6.534100E+01  0          0          0          0          0
awc         pctchg                -6.256550E+01  0          0          0          0          0
esco        pctchg                 3.305500E+01  0          0          0          0          0
perco       pctchg                 1.434000E+00  0          0          0          0          0
cn3_swf     pctchg                -7.710000E-02  0          0          0          0          0
canmx       pctchg                -6.680230E+01  0          0          0          0          0
lat_len     pctchg                -6.241200E+00  0          0          0          0          0
revap_co    pctchg                -5.854530E+01  0          0          0          0          0
surlag      pctchg                -2.125720E+01  0          0          0          0          0
```

In this file every record has `conds = 0` and `nspu = 0`, so each change applies to every HRU. The trailing integer count is `11` per row: `conds`, `lyr1`, `lyr2`, `year1`, `year2`, `day1`, `day2`, and `nspu` (eight integers), padded by the editor with extra zeros that the list-directed reader ignores.

Carbon parameters

SWAT+ wires 40 carbon-related parameters in `cal_parm_select.f90`. Each entry below maps a calibration name to its target field, suggests a working min/max range, and notes the object type (`bsn`, `lyr`, `plt`, `hru`) that must appear in the `ob_typ` column of `cal_parms.cal`. Bounds come from the sptAPI bundled `cal_parms.cal` (261 entries), with the temperature triple tightened to physical bounds to physically realistic ranges.

Basin scalars from `carbon.bsn` (18 entries)

name	ob_typ	units	abs_min	abs_max	description
init_seq	bsn	fraction	0.5	1.0	initial sequestration fraction (org_frac%frac_seq)
init_microb	bsn	fraction	0.0	0.1	initial microbial-humus fraction (org_frac%frac_hum_microb)
init_slow	bsn	fraction	0.2	0.6	initial slow-humus fraction (org_frac%frac_hum_slow)
init_passive	bsn	fraction	0.2	0.6	initial passive-humus fraction (org_frac%frac_hum_passive)
koc_c	bsn	none	500	1500	KOC for C transport (cb_wtr_coef%prmt_21)
solc_ratio	bsn	0..1	0.1	1.0	runoff:percolate soluble C ratio (cb_wtr_coef%prmt_44)
manure_c_frac	bsn	none	0.0	1.0	manure C partition between fresh and stable pools (man_coef%rtof)
bio_consol	bsn	none	0.05	0.5	biological consolidation factor (bio_consf)
till_consol	bsn	none	0.05	0.5	tillage consolidation factor (till_consf)
sfc_rsd_photodeg	bsn	1/day	0.0	0.05	surface residue photodegradation rate (photo_degrade_factor)
n_act_frac	bsn	fraction	0.005	0.04	fraction of organic N in active humus pool (n_act_frac)
cnr_cap	bsn	none	100	1000	upper cap on residue C:N ratio
cnr_ref	bsn	none	10	50	reference C:N ratio where decomp factor = 1
cpr_cap	bsn	none	1000	10000	upper cap on residue C:P ratio
cpr_ref	bsn	none	100	400	reference C:P ratio where decomp factor = 1
t_cbn_min	bsn	deg C	-5	5	minimum temperature bound for tmpf2 (org_con%tn); tightened to physical bounds
t_cbn_opt	bsn	deg C	25	40	optimum temperature for tmpf2 (org_con%top); tightened to physical bounds
t_cbn_max	bsn	deg C	45	55	maximum temperature bound for tmpf2 (org_con%tx); tightened to physical bounds

Note

The `min < opt < max` ordering invariant is not enforced by `cal_parms.cal` itself; calibration tools may still produce invalid combinations. The `fcgd` clamp ensures invalid orderings degrade gracefully (return 0) rather than crashing.

Per-layer fields from `carbon_lyr.bsn` (11 entries)

Each entry addresses `carbdb(ly)` or `org_allo(ly)` using the `lyr1`/`lyr2` slot of the `calibration.cal` record. Bounds are guarded to `[1, size(carbdb)]` inside the `case` branch, so out-of-range layer indices are silently no-op.

name	ob_typ	units	abs_min	abs_max	description
hp_rate	lyr	1/day	1.0e-06	1.0e-04	passive humus turnover rate (carbdb(ly)%hp_rate)
hs_rate	lyr	1/day	1.0e-05	1.0e-03	slow humus turnover rate (carbdb(ly)%hs_rate)
microb_rate	lyr	1/day	0.005	0.05	microbial biomass turnover rate (carbdb(ly)%microb_rate)
meta_rate	lyr	1/day	0.01	0.1	metabolic litter turnover rate (carbdb(ly)%meta_rate)
str_rate	lyr	1/day	0.005	0.05	structural litter turnover rate (carbdb(ly)%str_rate)
microb_top_rate	lyr	1/day	0.005	0.05	top-layer microbial activity adjustment (carbdb(ly)%microb_top_rate)
hs_hp	lyr	none	0.01	0.2	slow-to-passive humus allocation (carbdb(ly)%hs_hp)
a1co2	lyr	fraction	0.3	0.8	metabolic litter to CO2 (org_allo(ly)%a1co2)
asco2	lyr	fraction	0.3	0.8	slow humus to CO2 (org_allo(ly)%asco2)
apco2	lyr	fraction	0.3	0.8	passive humus to CO2 (org_allo(ly)%apco2)
abco2	lyr	fraction	0.0	0.8	microbial biomass to CO2 (org_allo(ly)%abco2)

Biomixing and tillage curve coefficients (6 entries)

Curve fit coefficients used by `mgt_tillfactor.f90` tillf option 4. The `zz_` prefix that the older code carried was dropped to match the column names in `carbon.bsn`.

name	ob_typ	units	abs_min	abs_max	description
bmix_a	bsn	none	1.0	5.0	biological mixing coefficient A
bmix_b	bsn	none	1.0	10.0	biological mixing coefficient B
bmix_c	bsn	none	-10.0	-1.0	biological mixing coefficient C
tillmix_a	bsn	none	1.0	5.0	tillage mixing coefficient A
tillmix_b	bsn	none	5.0	25.0	tillage mixing coefficient B
tillmix_c	bsn	none	-10.0	-1.0	tillage mixing coefficient C

Plant carbon parameters (5 entries)

Carbon-relevant parameters on the plant database (`pldb(ielem)`). These were not previously exposed even though plant biomass drives the whole NPP and residue chain.

name	ob_typ	units	abs_min	abs_max	description
bio_e	plt	(kg/ha)/(MJ/m^2)	10.0	60.0	biomass-energy ratio (drives NPP)
hvsti	plt	fraction	0.1	0.8	harvest index (controls residue vs grain C split)
rdmx	plt	m	0.5	3.0	maximum root depth
t_opt	plt	deg C	15	35	optimum plant growth temperature
t_base	plt	deg C	0	15	base plant growth temperature

The existing `usle_c` plant parameter is retained from earlier releases.

> **Note**
>
> For the `init_*` parameters to take effect, `proc_cal` must run **before** `soil_nutcarb_init`. The current `main.f90` already orders these correctly; do not reorder. Previously the order was reversed and `init_*` calibration was a silent no-op because the pools were already sized using pre-cal values.

Related files

- `carbon.bsn`. Basin-wide carbon scalar inputs whose fields these parameters target.
- `carbon_lyr.bsn`. Per-layer carbon inputs.
- Hard calibration. How these files are consumed during a calibration run.
- Conditions and regionalization. Condition-line vocabulary.